DHANYASHREE M V

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PROFESSIONAL SUMMARY

Final-year B.E. (AI & ML) graduate from DSCE Bengaluru (CGPA 9.0/10) with hands-on experience building and deploying cloud-integrated AI/ML solutions across the full lifecycle from data analysis and model development to production deployment via FastAPI, Docker, and MLflow. Skilled in Python automation, REST API development, and designing scalable pipelines on AWS, Azure, and GCP. Built end-to-end projects spanning multi-agent RAG systems, computer vision, and predictive analytics served on cloud infrastructure with real-time capabilities. Google Cloud certified, with a strong drive to continuously upskill across Cloud, DevOps, and Applied AI technologies.

EDUCATION

B.E. in Artificial Intelligence & Machine Learning *2022 – 2026*

Dayananda Sagar College of Engineering, Bengaluru | CGPA: 9.0 / 10

TECHNICAL SKILLS

Programming	Python, SQL, JavaScript, HTML, CSS
MLOps	MLflow (experiment tracking, model registry), Docker, Kubernetes, FastAPI, REST APIs, CI/CD basics, Vercel
AI/ML	LangChain, LangGraph, LLMs, RAG, Hugging Face, Scikit-learn, YOLOv11, AI Agents
Data Analysis	Pandas, NumPy, Matplotlib, Seaborn, Plotly, EDA, feature engineering, Tableau, Power BI
Databases	PostgreSQL, MongoDB, SQLite, SQLAlchemy
Tools & VCS	Git, GitHub, Streamlit, Flask, Label Studio, MLflow UI

PROFESSIONAL EXPERIENCE

Research & ML Intern - CSIR : National Aerospace Laboratories (NAL)

- Developed a data-processing workflow for aircraft structural health monitoring using strain signal data.
- Implemented ASTM E1049-85 Rainflow counting with exponential smoothing in Python; tracked algorithm performance across parameter variations to improve fatigue cycle accuracy.
- Generated and validated fatigue spectra for reliable RUL estimation maintained reproducible, version-controlled scripts and structured analysis documentation throughout.
- Conducted exploratory data analysis on strain signal datasets to identify trends, noise patterns, and anomalies before model input.

Data Science Intern - The Developers Arena (Remote)

- Built an end-to-end data science workflow: data cleaning, EDA (Pandas, Matplotlib, Seaborn), feature engineering, and ML model development using Python.
- Designed a production ML pipeline using Random Forest with preprocessing and served real-time predictions via a FastAPI microservice tracked model versions and performance metrics using MLflow.
- Built an interactive Airbnb Market Explorer Dashboard (Streamlit + Plotly) with KPI cards, filters, and geospatial price maps, surfacing pricing and neighbourhood trends from raw data.

PROJECTS

Cortex — Multi-Agent RAG System | LangGraph · LangChain · FastAPI · PostgreSQL · ChromaDB

- Built a 7-agent RAG pipeline for academic departmental assistance achieving 93.3% query success and ~3% hallucination rate across 300+ test queries.
- Integrated FastAPI backend with a React frontend for real-time query processing,connected PostgreSQL via SQLAlchemy to automate accreditation workflows.
- Tracked agent performance across iterative builds using MLflow maintained experiment logs to compare retrieval strategies and chunk configurations.

WildGuard AI | YOLOv11 · Python · Streamlit · Twilio API · Label Studio

- Developed a YOLOv11-based object detection system to identify poachers, rangers, and tourists from forest camera trap images.

- Annotated a custom dataset using Label Studio; tracked training runs and mAP scores across model versions using MLflow to select the best-performing checkpoint.
- Built a Streamlit UI supporting multi-image uploads and real-time predictions; integrated Twilio API for instant SMS alerts on poacher detection.

Airbnb Smart Pricing – Insights Hub | Python · Scikit-learn · FastAPI · Streamlit · MLflow

- Built a full-stack analytics platform combining EDA, interactive dashboards, and ML-based price prediction (Random Forest pipeline with preprocessing).
- Used MLflow for experiment tracking logged hyperparameters, RMSE, and R^2 scores across training runs to identify the optimal model configuration.
- Performed in-depth EDA using Pandas, Seaborn, and Plotly to surface pricing trends, neighbourhood insights, and room-type patterns from 50K+ listings.
- Served real-time predictions via FastAPI microservice consumed through REST APIs; deployed dashboard on Streamlit Cloud.

CERTIFICATIONS

- Google Cloud: Advanced - Generative AI for Developers Learning Path & Machine Learning
- Google Cloud Career Launchpad – Cloud Engineer Track(Vertex AI)
- NPTEL – Human Computer Interaction, IIT Madras

LEADERSHIP & ACTIVITIES

- Content Co-Lead, Aventus 3.0 Hackathon (National-Level) – DSCE Central Committee: Led cross-functional team coordination, deliverable planning, and quality review of all communications.
- Event Organiser, Innovation & Entrepreneurship Development Cell (IEDC), DSCE.